L2 Linguagem com construções imperativas (L2)

Sintaxe abstrata:

$$
\begin{array}{lcl}
e & \in & \text{Expr} \\
e & ::= & n \mid b \mid e_1\ op\ e_2 \mid \mathsf{if}\ e_1\ \mathsf{then}\ \{e_2\}\ \mathsf{else}\ \{e_3\} \\
 & \mid & x \mid \mathsf{let}\ x : T = e_1\ \mathsf{in}\ \{e_2\} \\
 & \mid & x := e_2 \mid\ !\, e \mid \mathsf{new}\ e \mid () \mid \mathsf{while}\ e_1\ \mathsf{do}\ \{e_2\} \mid e_1; e_2 \mid \boxed{\mathrm{l}}
\end{array}
$$

$$
\begin{array}{lcl}
v & \in & \text{Values} \\
v & ::= & n \mid b \mid () \mid \boxed{\mathrm{l}}
\end{array}
$$

$$
\begin{array}{lcl}
T & \in & \text{Types} \\
T & ::= & \mathsf{int} \mid \mathsf{bool} \mid \mathsf{ref}\ T \mid \mathsf{unit}
\end{array}
$$

$$
\begin{array}{lcll}
n & \in & \mathbb{Z} & \\
b & \in & \{\mathsf{true}, \mathsf{false}\} & \\
op & \in & \{+, <, \ldots\} & \text{(operações binárias, no mínimo adição e menor-que)} \\
\boxed{\mathrm{l}} & \in & \text{Locations} & \text{(localizações/endereços de memória)}
\end{array}
$$

Semântica operacional *small-step*:

$$\frac{e_1, \sigma \longrightarrow e_1', \sigma'}{e_1\ op\ e_2, \sigma \longrightarrow e_1'\ op\ e_2, \sigma'} \quad (\textsc{op}1) \qquad \frac{e_2, \sigma \longrightarrow e_2', \sigma'}{v\ op\ e_2, \sigma \longrightarrow v\ op\ e_2', \sigma'} \quad (\textsc{op}2)$$

$$\frac{[\![n]\!] = [\![n_1]\!] + [\![n_2]\!]}{n_1 + n_2, \sigma \longrightarrow n, \sigma} \quad (\textsc{op}+) \qquad \frac{[\![b]\!] = [\![n_1]\!] < [\![n_2]\!]}{n_1 < n_2, \sigma \longrightarrow b, \sigma} \quad (\textsc{op}<\textsc{b})$$

$$\mathsf{if\ true\ then}\ e_2\ \mathsf{else}\ e_3, \sigma \longrightarrow e_2, \sigma \quad (\textsc{if}1) \qquad \mathsf{if\ false\ then}\ e_2\ \mathsf{else}\ e_3, \sigma \longrightarrow e_3, \sigma \quad (\textsc{if}2)$$

$$\frac{e_1, \sigma \longrightarrow e_1', \sigma'}{\mathsf{if}\ e_1\ \mathsf{then}\ e_2\ \mathsf{else}\ e_3, \sigma \longrightarrow \mathsf{if}\ e_1'\ \mathsf{then}\ e_2\ \mathsf{else}\ e_3, \sigma'} \quad (\textsc{if}3)$$

$$\frac{e_1, \sigma \longrightarrow e_1', \sigma'}{\mathsf{let}\ x : T = e_1\ \mathsf{in}\ e_2, \sigma \longrightarrow \mathsf{let}\ x : T = e_1'\ \mathsf{in}\ e_2, \sigma'} \quad (\textsc{e-let}1)$$

$$\frac{}{\mathsf{let}\ x : T = v\ \mathsf{in}\ e_2, \sigma \longrightarrow \{v/x\}\ e_2, \sigma} \quad (\textsc{e-let}2)$$

$$\frac{l \in Dom(\sigma)}{l := v, \sigma \longrightarrow (), \sigma[l \mapsto v]} \quad (\textsc{atr}1) \qquad \frac{e, \sigma \longrightarrow e', \sigma'}{l := e, \sigma \longrightarrow l := e', \sigma'} \quad (\textsc{atr}2)$$

$$\frac{l \in Dom(\sigma) \quad \sigma(l) = v}{!\, l, \sigma \longrightarrow v, \sigma} \quad (\textsc{deref}1) \qquad \frac{e, \sigma \longrightarrow e', \sigma'}{!\, e, \sigma \longrightarrow !\, e', \sigma'} \quad (\textsc{deref})$$

$$\frac{l \notin Dom(\sigma)}{\mathsf{new}\ v, \sigma \longrightarrow l, \sigma[l \mapsto v]} \quad (\textsc{new}1) \qquad \frac{e, \sigma \longrightarrow e', \sigma'}{\mathsf{new}\ e, \sigma \longrightarrow \mathsf{new}\ e', \sigma'} \quad (\textsc{new})$$

$$\frac{}{(); e_2, \sigma \longrightarrow e_2, \sigma} \quad (\textsc{seq}1) \qquad \frac{e_1, \sigma \longrightarrow e_1', \sigma'}{e_1; e_2, \sigma \longrightarrow e_1'; e_2, \sigma'} \quad (\textsc{seq})$$

$$\mathsf{while}\ e_1\ \mathsf{do}\ e_2, \sigma \longrightarrow \mathsf{if}\ e_1\ \mathsf{then}\ (e_2; \mathsf{while}\ e_1\ \mathsf{do}\ e_2)\ \mathsf{else}\ (), \sigma \quad (\textsc{e-while})$$

Sistema de Tipos:

$$\frac{}{\Gamma \vdash n : \mathsf{int}} \quad (\text{T-INT})$$

$$\frac{}{\Gamma \vdash b : \mathsf{bool}} \quad (\text{T-BOOL})$$

$$\frac{\Gamma \vdash e_1 : \mathsf{int} \qquad \Gamma \vdash e_2 : \mathsf{int}}{\Gamma \vdash e_1 + e_2 : \mathsf{int}} \quad (\text{T-OP}+)$$

$$\frac{\Gamma \vdash e_1 : \mathsf{int} \qquad \Gamma \vdash e_2 : \mathsf{int}}{\Gamma \vdash e_1 < e_2 : \mathsf{bool}} \quad (\text{T-OP}<)$$

$$\frac{\Gamma \vdash e_1 : \mathsf{bool} \qquad \Gamma \vdash e_2 : T \qquad \Gamma \vdash e_3 : T}{\Gamma \vdash \mathsf{if}\ e_1\ \mathsf{then}\ e_2\ \mathsf{else}\ e_3 : T} \quad (\text{T-IF})$$

$$\frac{\Gamma(x) = T}{\Gamma \vdash x : T} \quad (\text{T-VAR})$$

$$\frac{\Gamma \vdash e_1 : T \qquad \Gamma, x \mapsto T \vdash e_2 : T'}{\Gamma \vdash \mathsf{let}\ x : T = e_1\ \mathsf{in}\ e_2 : T'} \quad (\text{T-LET})$$

$$\frac{\Gamma(x) = \mathsf{ref}\ T \qquad \Gamma \vdash e_2 : T}{\Gamma \vdash x := e_2 : \mathsf{unit}} \quad (\text{T-ATR})$$

$$\frac{\Gamma \vdash e : \mathsf{ref}\ T}{\Gamma \vdash\ !\, e : T} \quad (\text{T-DEREF})$$

$$\frac{\Gamma \vdash e : T}{\Gamma \vdash \mathsf{new}\ e : \mathsf{ref}\ T} \quad (\text{T-NEW})$$

$$\frac{}{\Gamma \vdash () : \mathsf{unit}} \quad (\text{T-UNIT})$$

$$\frac{\Gamma \vdash e_1 : \mathsf{bool} \qquad \Gamma \vdash e_2 : \mathsf{unit}}{\Gamma \vdash \mathsf{while}\ e_1\ \mathsf{do}\ e_2 : \mathsf{unit}} \quad (\text{T-WHILE})$$

$$\frac{\Gamma \vdash e_1 : \mathsf{unit} \qquad \Gamma \vdash e_2 : T}{\Gamma \vdash e_1;\ e_2 : T} \quad (\text{T-SEQ})$$

Trabalho

O trabalho consiste em

- parser,
- inferência de tipos e
- avaliador *big step*

O trabalho será avaliado da seguinte forma:

- nota máxima 6,0 para os trabalhos que implementarem somente a inferência de tipos OU somente o avaliador *big step* para L2 conforme a especificação dada acima
- nota máxima 7,5 para os trabalhos que implementarem (i) a inferência de tipos E o avaliador *big step* OU (ii) a inferência de tipos E parser, OU (iii) avaliador *big step* E parser
- nota máxima 10,0 para os trabalhos que implementarem o parser, a inferência de tipos, e o avaliador *big step*

O trabalho deve ser realizado em grupos de 3 componentes, e de ser entregue via Moodle no prazo especificado. Após a entrega, o trabalho será apresentado em laboratório pelos componentes do grupo, conforme cronograma de apresentações disponível no Moodle.